Integrated Science **Name** _____
Unit 3: Matter, Energy & Atmosphere **Date:** _____**Per:** _____

Formula Worksheet

Learning goal: To understand the information a chemical formula provides.

In order to study the atmosphere in further depth, you must understand a little more chemistry. This worksheet is designed to help you understand what is meant by a chemical formula.

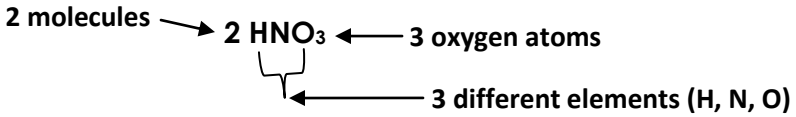
One of the main components of 'acid rain' is nitric acid, **2 HNO_3**. Let's examine this molecule a little closer:

There are **two molecules** represented here. The *coefficient,* or the number in front of the molecule, is two. If there is no coefficient, then it is assumed there is one of the molecule.

There are **3 different *elements*** in this one molecule: Hydrogen, Nitrogen and Oxygen. Each new element starts with a capital letter. If an element has two letters in its symbol, the second is lower cased; for example, sodium, Na.

There is a total of **5 atoms** in this molecule. This is determined by the *subscripts* to the lower right of each element. There is one Hydrogen (it is assumed to be one if there is NO subscript), one Nitrogen and three Oxygens. If there is a subscript outside a set of parenthesis, that subscript distributes (is multiplied by) to each atom inside the parenthesis.

There are a **total of ten atoms** in these two molecules of HNO_3. This can be determined by multiplying the coefficient by each subscript. There are two hydrogen atoms, two nitrogen atoms and six oxygen atoms. This adds up to ten total atoms.



	Chemical Formula	Number of Molecules	Number of Elements	Number of Atoms per Molecule	Total number of Atoms present
Example	**$3H_2O$**	**3**	**2**	**3**	**9**
1.	**4NaCl**				
2.	**$2CaCl_2$**				
3.	**H_2SO_4**				
4	**Na_3PO_4**				
5.	**$2H_2O_2$**				

	Chemical Formula	Number of Molecules	Number of Elements	Number of Atoms per Molecule	Total number of Atoms present
6.	$3C_6H_6$				
7.	$2C_7H_{16}$				
8.	$5AgBr$				
9.	NH_4Cl				
10.	$KHCO_3$				
11.	$3K_2CrO_3$				
12.	$C_2H_2Cl_4$				
13.	$2CoCl_2$				
14.	$3BaBr_2$				
15.	$Ca_3(PO_4)_2$				